STALE 2027: Workshop Proposal for IEEE INFOCOM 2027

1. Title

STALE 2027. First International Workshop on Acting on Stale State in Networked Systems

Short name: STALE. Half day, in person, Honolulu, May 24, 2027.

2. Organization team

- Andrew H. Bond, San Jose State University, USA (General Chair)
- [TPC Co-Chair, second institution, ideally outside the US] TBD
- [TPC Co-Chair or Challenge Chair, ideally a cluster operator or national research platform staff member] TBD
- [Publicity Chair, ideally an early-career researcher] TBD

3. Scope and topics

Every networked controller acts on a picture of the world that was true some time ago. A cluster admission gate reads a capacity record that was sampled by one process, carried by a message bus, and consulted after a scheduler has already moved on. A congestion controller reacts to signals that are one round trip old. A spectrum user senses a channel, then transmits into whatever the channel has become. Edge orchestrators, federated learning coordinators, and digital twins all share this problem. The decision is made now. The evidence is from the past. The consequences land on someone else.

STALE brings together researchers who study how old information should change what a networked system does. We are especially interested in work that connects three kinds of evidence that are usually published separately. The first is a model that says what the best possible policy can achieve for a given information age. The second is a finite-sample statement about a policy fitted from real data. The third is a deployed system whose records let an independent reader check what actually happened.

Topics include, but are not limited to, the following.

- Age of information and value of information in control loops, admission control, and scheduling
- Admission and offloading decisions driven by delayed or partial capacity records
- Guest-side behavior on shared infrastructure, including opportunistic and volunteer clusters, research platforms, and multi-tenant GPU pools
- Limits on what feedback can achieve when observations are delayed, and when a static policy is as good as an adaptive one
- Budgeted and risk-constrained decisions under delayed outcome labels
- Message-bus and publish/subscribe control planes, including leaf federation, delivery semantics, and their effect on decision freshness

- Cloud and edge bursting, cold start, warm pools, and the cost of acting late
- Learned controllers (reinforcement learning, bandits) under stale observations, and tests of whether they beat simple baselines
- Opportunistic spectrum access and collision-constrained access with aged sensing
- Measurement studies of staleness in production systems, including Kubernetes, Slurm, and cloud control planes
- Audit, provenance, and replay of control decisions from append-only event records
- Benchmarks, traces, and simulators for stale-state control

We welcome regular papers (up to 6 pages), short papers on early results or negative results (up to 4 pages), and demo papers (up to 3 pages plus a live demo). Negative results are explicitly in scope. A paper showing that a feedback controller cannot beat a static budget under realistic delay is as useful as one showing that it can.

4. Rationale

4a. Relevance to the INFOCOM community

Information freshness is a long-standing INFOCOM topic. Age of information, delayed feedback in congestion control, opportunistic spectrum access, and scheduling under partial observation have all appeared in the main program for more than a decade. Most of that work stops at the model. STALE asks what happens when these models meet real control planes, where the age of a record is itself a measured, noisy quantity and the outcome of a decision arrives late or not at all. That connection between networking theory and operated systems is the core of INFOCOM's identity.

4b. Timeliness

Three changes make the topic urgent now.

1. Shared GPU capacity is scarce and is increasingly accessed by guests who do not control placement. National research platforms, campus clusters, and spot markets all rely on users to behave well with whatever partial view of the cluster they have. Policies that ban or throttle users who over-request make guest-side decisions consequential.
2. Control planes are becoming message-driven. Systems built on NATS, Kafka, and similar buses decouple the process that samples state from the process that acts on it. That decoupling adds delay that existing models rarely account for.
3. Learned controllers are being deployed for scheduling and admission. Whether they provide any benefit when observations are old is an open empirical question with direct cost implications.

4c. Expected submissions

The topic sits where several active communities overlap. Age of information, cognitive radio, and network control theory supply the models. Systems and HPC researchers supply deployments and traces. Machine learning for systems supplies learned controllers. Each of these groups currently publishes this kind of work as a side result inside papers about something else. A dedicated venue with a shared benchmark gives them a reason to write it up directly. We target 20 to 30 submissions and 8 to 10 accepted papers for the first edition.

The short submission window (paper deadline December 29, 2026) favors early results and short papers, which matches the workshop's purpose. The open benchmark described below also lowers the cost of producing a credible submission.

4d. Attracting attendees

- **The Stale-State Challenge.** We will release, at the time of the call, an open replay harness and a set of capacity traces with known delays. Entrants submit an admission policy. The harness scores each policy on the work it gets done against a fixed budget for decisions made while capacity was unavailable, and checks every decision against the recorded event log. The leaderboard and a short talk by the top entries close the workshop. A challenge gives students a concrete entry point and gives attendees something to follow.
- **A live demo session.** Demos run against a real message-bus control plane that bursts Kubernetes Jobs onto a shared research cluster from a laptop in the room, showing record age, decisions, and outcomes on a live dashboard.
- **A panel of operators and researchers** titled "What a good guest looks like." Cluster operators state what they need from guest workloads, and researchers say what their models can and cannot promise.
- **Publicity** through the INFOCOM, MobiHoc, and SIGMETRICS mailing lists, research platform user communities, and NATS and Kubernetes community channels.

5. Tentative committees

Organizing committee. See Section 2.

Technical program committee (to be confirmed, target 15 to 20 members across age of information, network control, cognitive radio, systems and HPC, and machine learning for systems).

- [Name, affiliation] TBD
- [Name, affiliation] TBD
- ...

Organizers will not handle their own submissions. Any paper with an organizer as author will be managed by a TPC member without a conflict, and organizer papers will not exceed two.

6. Draft call for papers

> **STALE 2027. First International Workshop on Acting on Stale State in Networked Systems.** In conjunction with IEEE INFOCOM 2027, Honolulu, Hawaii, USA, May 24, 2027.
>
> Every networked controller decides now using information from the past. Admission gates read capacity records that were sampled seconds ago and carried over a message bus. Congestion controllers react to signals one round trip old. Spectrum users transmit into channels they sensed earlier. STALE invites work on how old information should change what a networked system does, and on how to show, with models, data, and auditable deployments, that it does.
>
> Topics of interest include age and value of information in control loops, admission and offloading from delayed capacity records, guest-side behavior on shared clusters, limits of feedback under delay, budgeted decisions under delayed outcome labels, message-bus control planes, cloud and edge bursting, learned controllers under stale observations, opportunistic spectrum access with aged sensing, measurement studies of staleness in production systems, audit and replay of control decisions, and benchmarks and traces for stale-state control.
>
> Negative results are welcome.

Stale-State Challenge. An open replay harness and capacity traces are available at https://github.com/ahb-sjsu/stale-bench. Challenge entries are submitted as a 2-page description plus code and are eligible for presentation in the challenge session.

Submission formats. Regular papers up to 6 pages, short papers up to 4 pages, demo papers up to 3 pages, all in IEEE two-column conference format including references. Reviews are double-blind. Accepted papers will be submitted for inclusion in IEEE Xplore. At least one author of each accepted paper must register and present in person.

Important dates. Paper submission December 29, 2026. Notification February 2, 2027. Camera ready February 16, 2027. Workshop May 24, 2027.

7. Schedule

Milestone	Date
Call for papers and challenge harness released	November 11, 2026
a. Paper submission deadline	December 29, 2026
b. Review deadline (proposed)	January 22, 2027
c. Acceptance notification	February 2, 2027
d. Camera ready	February 16, 2027
e. Program ready (set by INFOCOM and IEEE)	March 4, 2027
Challenge leaderboard frozen	April 30, 2027
Workshop	May 24, 2027

9. Format

Half day, two sessions of 90 minutes, sized for 8 to 10 accepted papers.

- Session 1. Opening (5 min). Invited talk (30 min). Four paper talks (55 min).
- Session 2. Four to six paper talks and demos (50 min). Challenge results (15 min). Panel "What a good guest looks like" (25 min).

If submissions substantially exceed expectations we will ask the workshop chairs about a full day.

10. Keynote and panel

Yes. One invited talk from a researcher in age of information or network control theory [name TBD, to be confirmed before the proposal is submitted if possible], and one panel of three to four speakers mixing research cluster operators and researchers.

11. Workshop history

This is the first edition.